

Front-end Interview Questions

HTML

1. What is HTML and what are its uses?
2. Is HTML the only language used to create web pages?
3. What is the purpose of `<!DOCTYPE html>`?
4. What is a boilerplate in HTML?
5. What are meta tags?
6. What are semantic elements in HTML?
7. What is the difference between `<div>` and `<span>`?
8. What is an attribute in HTML?
9. What are the two types of tags in HTML?
10. How do you add a link in a web page?
11. What is the difference between `href` and `src`?
12. What are the attributes of the `<img>` tag?
13. What are forms in HTML?
14. What is the difference between label and placeholder?
15. What is the difference between HTML4 and HTML5?

CSS

1. What is CSS and why do we use it?
2. What are the different ways to apply CSS?
3. What is the difference between class and id selectors?

4. What is CSS specificity? Explain the order.
5. Explain the CSS Box Model.
6. What is the difference between margin and padding?
7. What is the difference between display: none and visibility: hidden?
8. Explain CSS positioning and its types.
9. What is Flexbox?
10. How do you center a div?
11. What is CSS Grid?
12. What is the difference between Flexbox and Grid?
13. What is responsive web design?
14. What are media queries and how do they work?
15. What are the different CSS units used for responsive design?

JavaScript

1. What is JavaScript and why is it used?
2. What are the different data types in JavaScript?
3. What is the difference between var, let, and const?
4. What is hoisting in JavaScript?
5. What is scope? Explain global, function, and block scope.
6. What is the difference between == and ===?
7. What is a closure in JavaScript?
8. What is the difference between null and undefined?
9. What are truthy and falsy values?
10. What is an arrow function? How is it different from a normal function?
11. What is the this keyword in JavaScript?

12. What is an Immediately Invoked Function Expression (IIFE)?
13. What are callbacks in JavaScript?
14. What is the difference between synchronous and asynchronous JavaScript?
15. What is a Promise?
16. What is async and await?
17. What is the event loop in JavaScript?
18. What is the difference between map(), filter(), and reduce()?
19. What is event delegation?
20. What is DOM manipulation?

React

1. What is React and why is it used?
2. What are the main features of React?
3. What is JSX?
4. What is the difference between JSX and HTML?
5. What is a component in React?
6. What are the types of components in React?
7. What are props in React?
8. What is state in React?
9. What is the difference between props and state?
10. What is the virtual DOM?
11. How does React update the DOM efficiently?
12. What are hooks in React?
13. What is the useState hook?
14. What is the useEffect hook?

15. What is conditional rendering in React?
16. What is list rendering and why are keys important?
17. What is lifting state up?
18. What is controlled vs uncontrolled components?
19. What is React Router and why is it used?
20. What is the difference between class components and functional components?

Vue.js

1. What is Vue.js and why is it used?
2. What are the main features of Vue.js?
3. What is a Vue instance?
4. What is template syntax in Vue?
5. What are directives in Vue?
6. What is data binding in Vue?
7. What is reactivity in Vue?
8. What are computed properties?
9. What are watchers in Vue?
10. What are components in Vue?
11. What is props in Vue?
12. What is event handling in Vue?
13. What is conditional rendering in Vue?
14. What is list rendering in Vue?
15. What is Vue Router and why is it used?

Angular

1. What is Angular?
2. What is the difference between AngularJS and Angular?
3. What are components in Angular?
4. What is a module in Angular?
5. What are directives in Angular?
6. What is data binding in Angular?
7. What are services in Angular?
8. What is dependency injection?
9. What is routing in Angular?
10. What are pipes in Angular?
11. What is lifecycle hook in Angular?
12. What is ngIf and ngFor?
13. What is observables in Angular?
14. What is HttpClient in Angular?
15. What is the difference between template-driven and reactive forms?

SCSS

1. What is SCSS and why do we use it?
2. What is the difference between CSS and SCSS?
3. What are variables in SCSS?
4. What are mixins in SCSS?
5. What is the difference between @use and @import?
6. What is nesting in SCSS?
7. What are partials in SCSS?

8. What are functions in SCSS?
9. What are operators in SCSS?
10. How is SCSS compiled into CSS?

Tailwind CSS

1. What is Tailwind CSS?
2. How is Tailwind different from traditional CSS frameworks?
3. What are utility-first classes?
4. What is the purpose of `tailwind.config.js`?
5. How do you customize colors and fonts in Tailwind?
6. What is responsive design in Tailwind?
7. What are variants in Tailwind?
8. What is the difference between `@apply` and utility classes?
9. How does Tailwind handle dark mode?
10. What are the advantages and disadvantages of Tailwind CSS?

Bootstrap

1. How does Bootstrap help in building responsive layouts?
2. How does the Bootstrap grid system work internally?
3. What is the purpose of `container`, `row`, and `col` in Bootstrap?
4. How do Bootstrap breakpoints work?
5. How does Bootstrap make a layout responsive without writing custom media queries?
6. How do utility classes work in Bootstrap?
7. How does Bootstrap handle spacing like margin and padding?

8. How do Bootstrap components work (buttons, cards, navbar)?
9. How does Bootstrap JavaScript work for components like modal and dropdown?
10. What is the role of data attributes in Bootstrap?
11. How does Bootstrap ensure cross-browser compatibility?
12. How do you customize Bootstrap styles in a project?
13. How do you override Bootstrap default styles safely?
14. How does Bootstrap handle mobile-first design?
15. What are the common mistakes developers make while using Bootstrap?

Web Fundamentals

1. What is the client-server model?
2. What happens when you type a URL in the browser?
3. What is HTTP and HTTPS?
4. What are HTTP methods?
5. What is the difference between GET and POST?
6. What are HTTP status codes?
7. What is CORS?
8. What is caching in the browser?
9. What is session vs cookie?
10. What is localStorage vs sessionStorage?

API Concepts

1. What is an API?

2. What is REST API?
3. What are RESTful principles?
4. What is JSON?
5. What is the difference between REST and SOAP?
6. What is authentication vs authorization?
7. What is JWT?
8. What is pagination in APIs?
9. What is rate limiting?
10. What is API versioning?

git and GitHub

1. What is Git?
2. Why do we use Git?
3. What is a version control system?
4. What is the difference between Git and other version control systems?
5. What is a repository in Git?
6. What is the difference between a local repository and a remote repository?
7. What is git init?
8. What is git clone?
9. What is the difference between git add and git commit?
10. What is the purpose of git status?
11. What is the difference between git pull and git fetch?
12. What is a branch in Git?
13. Why do we use branches?
14. What is git merge?

15. What is a merge conflict?
16. What is GitHub?
17. What is the difference between Git and GitHub?
18. What is a pull request?
19. What is a fork in GitHub?
20. What is a README file and why is it important?
21. What is .gitignore?
22. How do you collaborate with a team using GitHub?
23. What is the difference between public and private repositories?
24. What is commit history?
25. What are issues in GitHub?
26. How do you push code to GitHub for the first time?
27. How do you update your local code from GitHub?
28. What steps do you follow to resolve a merge conflict?
29. How do you create and switch branches?
30. What are the common Git commands you use daily?